Using Logic

Welcome to the Purdue OWL

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Logical Vocabulary

Before using logic to reach conclusions, it is helpful to know some important vocabulary related to logic.

Premise: Proposition used as evidence in an argument.

Conclusion: Logical result of the relationship between the premises. Conclusions serve as the thesis of the argument.

Argument: The assertion of a conclusion based on logical premises.

Syllogism: The simplest sequence of logical premises and conclusions, devised by Aristotle.

Enthymeme: A shortened syllogism which omits the first premise, allowing the audience to fill it in. For example, "Socrates is mortal because he is a human" is an enthymeme which leaves out the premise "All humans are mortal."

Induction: A process through which the premises provide some basis for the conclusion.

Deduction: A process through which the premises provide conclusive proof for the conclusion.

Reaching Logical Conclusions

Reaching logical conclusions depends on the proper analysis of premises. The goal of a syllogism is to arrange premises so that only one true conclusion is possible.

Example A:

Consider the following premises:

> **Premise 1:** Non-renewable resources do not exist in infinite supply.
> **Premise 2:** Coal is a non-renewable resource.

From these two premises, only one logical conclusion is available:

> **Conclusion:** Coal does not exist in infinite supply.

Example B:

Often logic requires several premises to reach a conclusion.

Premise 1: All monkeys are primates.

Premise 2: All primates are mammals.

Premise 3: All mammals are vertebrate animals. **Conclusions:** Monkeys are vertebrate animals.

Example C:

Logic allows specific conclusions to be drawn from general premises. Consider the following premises:

Premise 1: All squares are rectangles.

Premise 2: Figure 1 is a square.

Conclusion: Figure 1 is also a rectangle.

Syllogistic Fallacies

The syllogism is a helpful tool for organizing persuasive logical arguments. However, if used carelessly, syllogisms can instill a false sense of confidence in unfounded conclusions. The examples in this section demonstrate how this can happen.

Example D:

Logic requires decisive statements in order to work. Therefore, this syllogism is false:

Premise 1: Some quadrilaterals are squares.

Premise 2: Figure 1 is a quadrilateral.

Conclusion: Figure 1 is a square.

This syllogism is false because not enough information is provided to allow a verifiable conclusion. Figure 1 could just as likely be a rectangle, which is also a quadrilateral.

Example E:

Logic can also mislead when it is based on premises that an audience does not accept. For instance:

> **Premise 1:** People with red hair are not good at checkers.
> **Premise 2:** Bill has red hair.
> **Conclusion:** Bill is not good at checkers.

Within the syllogism, the conclusion is logically valid. However, the syllogism itself is only true if an audience accepts Premise 1, which is very unlikely. This is an example of how logical statements can appear accurate while being completely false.

Example F:

Logical conclusions also depend on which factors are recognized and ignored by the premises. Therefore, premises that are correct but that ignore other pertinent information can lead to incorrect conclusions.

> **Premise 1:** All birds lay eggs.
> **Premise 2:** Platypuses lay eggs.
> **Conclusion:** Platypuses are birds.

It is true that all birds lay eggs. However, it is also true that some animals that are not birds lay eggs. These include fish, amphibians, reptiles, and a small number of mammals (like the platypus and echidna). To put this another way: laying eggs is not a defining characteristic of birds. Thus, the syllogism, which assumes that because all birds lay eggs, *only* birds lay eggs, produces an incorrect conclusion.

A better syllogism might look like this:

> **Premise 1:** All mammals have fur.
> **Premise 2:** Platypuses have fur.

Conclusion: Platypuses are mammals.

Fur is indeed one of the defining characteristics of mammals—in other words, there are not non-mammal animals who also have fur. Thus, the conclusion here is more firmly-supported.

In sum, though logic is a very powerful argumentative tool and is far preferable to a disorganized argument, logic does have limitations. It must also be effectively developed from a syllogism into a written piece.